Engenharia de Software
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2ª Lista de Exercícios de Álgebra Linear

1ª questão

a) $\begin{bmatrix} 1 & 0 & 0 & -7 & 8 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & 1 & -5 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ $\begin{bmatrix} x \\ y \\ z \\ j \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ -5 \end{bmatrix}$ $\begin{bmatrix} 1_x & 0_y & 0_z & -7_j \\ 0_x & 1_y & 0_z & 3_j \\ 0_x & 0_y & 1_z & 1_j \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ -5 \end{bmatrix}$

$$\begin{cases} 1_x \; 0_y \; 0_z \; -7_j = 8 \\ 0_x \; 1_y \; 0_z \; 3_j = 2 \\ 0_x \; 0_y \; 1_z \; 1_j = -5 \end{cases} \qquad \begin{cases} 1_x = 8 + 7_j \\ 1_y = 2 - 3_j \\ 1_z = -5 - 1_j \end{cases}$$

//

b) $\begin{bmatrix} 1 & -6 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $\left[\begin{array}{ccccc|c} 1 & -6 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$

$x_4 + 5x_5 + 8$

$x_4 = 8 - 5x_5$

$x_3 + 0x_4 + 4x_5 = 7$

$x_3 = 7 - 4x_5$

$x - 6x_2 + 3x_5 = -2$

$x_1 - 6x_2 = -2 - 3x_5$

$$S = \begin{bmatrix} -2 - 3x_5 + 6x_2 \\ x_2 \\ 7 - 4x_5 \\ 8 - 5x_5 \\ x_5 \end{bmatrix}$$

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c) $\begin{bmatrix} 1 & 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ i \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix} \quad \begin{bmatrix} 1_x & 0_y & 0_z & 0_i \\ 0_x & 1_y & 0_z & 0_i \\ 0_x & 0_y & 1_z & 1_i \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$ 7x0

$\begin{cases} 1_x\ 0_y\ 0_z\ 0_i = 6 \\ 0_x\ 1_y\ 0_z\ 0_i = 3 \\ 0_x\ 0_y\ 1_z\ 1_i = 2 \end{cases} \quad \begin{cases} 1_x = 6 \\ 1_y = 3 \\ 1_z = 2 - 1_i \end{cases}$

d) $\begin{bmatrix} 1 & -6 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ ~ Igual a B -2

2° questão

a) $\begin{cases} x_1 + x_2 + 2x_3 = 8 \\ -x_1 - 2x_2 + 3x_3 = 1 \\ 3x_2 - 7x_2 + 4x_3 = 10 \end{cases}$

$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right] \begin{matrix} \times(1) \\ L_2 \to (-1)L_1 + L_2 \\ \end{matrix} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 3 & -7 & 4 & 10 \end{array}\right] \begin{matrix} \times(-3) \\ L_3 - 3 \cdot L_1 \to L_3 \end{matrix}$

$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right] \begin{matrix} \times(-1) \\ L_2/(-1) \to L_2 \end{matrix} \quad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & -5 & -9 \\ 0 & -10 & -2 & -14 \end{array}\right] \begin{matrix} \times(-10) \\ L_3 - (-10) \cdot L_2 \to L_3 \end{matrix} \quad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & -52 & -104 \end{array}\right] \begin{matrix} L_3/(-52) \to L_3 \\ \times(-1/52) \end{matrix}$

$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array}\right] \begin{matrix} L_2 - (-5) \cdot L_3 \to L_2 \\ \times(5) \end{matrix} \quad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \begin{matrix} L_1 - 2 \cdot L_3 \to L_1 \\ \times(-2) \end{matrix} \quad \left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \begin{matrix} \times(-1) \\ L_1 - L_2 \to L_1 \end{matrix} \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \begin{matrix} x_1 = 3 \\ x_2 = 1 \\ x_3 = 2 \end{matrix}$

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b) $\begin{cases} 2x_1 + 2x_2 + 2x_3 = 0 \\ -2x_1 + 5x_2 + 2x_3 = 1 \\ 8x_1 + x_2 + 4x_3 = -1 \end{cases}$

$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right] \times\left(\frac{1}{2}\right) \; L_1/(2) \to L_1 \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right] \times(2) \; L_2(-2)\cdot L_1 \to L_2$

$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right] \times(-8) \; L_3 - 8\cdot L_1 \to L_3 \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{array}\right] \times\left(\frac{1}{7}\right) \; L_2/(7) \to L_2 \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & -7 & -4 & -1 \end{array}\right] \times 7 \; L_3 - (-7)\cdot L_2 \to L_3$

$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 \end{array}\right] \times(-1) \; L_1 - 1\cdot L_2 \to L_1 \quad \left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{7} & \frac{-1}{7} \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 \end{array}\right]$

$\begin{cases} x_1 = \frac{-1}{7} - \frac{3}{7}\cdot x_3 \\ x_2 = \frac{1}{7} - \frac{4}{7}\cdot x_3 \\ x_3 = x_3 \end{cases} \quad X = \begin{bmatrix} \frac{-1}{7} - \frac{3}{7}\cdot x_3 \\ \frac{1}{7} - \frac{4}{7}\cdot x_3 \\ x_3 \end{bmatrix}$

c) $\begin{cases} -2x_2 + 3x_3 = 1 \\ 3x_1 + 6x_2 - 3x_3 = 1 \\ 6x_1 + 6x_2 + 3x_3 = 5 \end{cases}$

$\left[\begin{array}{ccc|c} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{array}\right] L_2 \leftrightarrow L_1 \quad \left[\begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array}\right] \times\left(\frac{1}{3}\right) \; L_1/3 \to L_1$

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & \frac{-2}{3} \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array}\right] \times(-6) \; L_3 - 6\cdot L_1 \to L_3 \quad \left[\begin{array}{ccc|c} 1 & 2 & -1 & \frac{-2}{3} \\ 0 & -2 & 3 & 1 \\ 0 & -6 & 9 & 9 \end{array}\right] \times\left(\frac{-1}{2}\right) \; L_2/(-2) \to L_2 \quad \left[\begin{array}{ccc|c} 1 & 2 & -1 & \frac{-2}{3} \\ 0 & 1 & \frac{-3}{2} & \frac{-1}{2} \\ 0 & -6 & 9 & 9 \end{array}\right] \times(6) \; L_3 - (-6)\cdot L_2 \to L_3$

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & \frac{-2}{3} \\ 0 & 1 & \frac{-3}{2} & \frac{-1}{2} \\ 0 & 0 & 0 & 6 \end{array}\right] = \begin{cases} x_1 + 2\cdot x_2 - x_3 = \frac{-2}{3} \\ x_2 - \frac{3}{2}\cdot x_3 = \frac{-1}{2} \\ = 6 \end{cases}$ Não tem solução

3ª questão - Os sistemas lineares seguintes possuem a mesma matriz A. Resolva-os usando o método de Gauss-Jordan. Observe que os dois sistemas podem ser resolvidos ao mesmo tempo escalonando a matriz aumentada $[A|B_1|B_2]$

a) $\begin{cases} x_1 - 2x_2 + x_3 = 1 \\ 2x_1 - 5x_2 + x_3 = -2 \\ 3x_1 - 7x_2 + 2x_3 = -1 \end{cases}$

$L_2 - 2L_1 \to L_2$

b) $\begin{cases} x_1 - 2x_2 + x_3 = 2 \\ 2x_1 - 5x_2 + x_3 = 1 \\ 3x_1 - 7x_2 + 2x_3 = 2 \end{cases}$

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a) $\begin{cases} x_1 - 2x_2 + x_3 = 1 \\ -x_2 - x_3 = -4 \\ 3x_1 - 7x_2 + 2x_3 = 1 \end{cases}$

$L_3 - 3L_1 \to L_3$

$\begin{cases} x_1 - 2x_2 + x_3 = 1 \\ -x_2 - x_3 = -4 \\ -x_2 - x_3 = -4 \end{cases}$

$-L_2 \to L_2$

$\begin{cases} x_1 - 2x_2 + x_3 = 1 \\ x_2 + x_3 = 4 \\ -x_2 - x_3 = -4 \end{cases}$

$L_3 + L_2 \to L_3$

$\begin{cases} x_1 - 2x_2 + x_3 = 1 \\ x_2 + x_3 = 4 \\ 0 = 0 \end{cases}$

$L_1 + 2L_2 \to L_1$

$\begin{cases} x_1 + 3x_3 = 9 \\ x_2 + x_3 = 4 \\ 0 = 0 \end{cases}$

$x_1 = 9 - 3x_3$

$x_2 = 4 - x_3$

$x_3 = x_3$ //

B) $L_2 - 2L_1 \to L_2$ $\quad L_3 - 3L_1 \to L_3$

$\begin{cases} x_1 - 2x_2 + x_3 = 2 \\ -x_2 - x_3 = 3 \\ 3x_1 - 7x_2 + 2x_3 = 2 \end{cases}$

$\begin{cases} x_1 - 2x_2 + x_3 = 2 \\ -x_2 - x_3 = 3 \\ -x_2 - x_3 = 4 \end{cases}$

$-L_2 \to L_2$

$\begin{cases} x_1 - 2x_2 + x_3 = 2 \\ x_2 + x_3 = 3 \\ -x_2 - x_3 = 4 \end{cases}$

$L_3 + L_2 \to L_3$

$\begin{cases} x_1 - 2x_2 + x_3 = 2 \\ x_2 + x_3 = 3 \\ 0 = 7 \end{cases}$

$L_1 + 2L_2 \to L_1$

$\begin{cases} x_1 + 3x_3 = 8 \\ x_2 + x_3 = 3 \\ 0 = 7 \end{cases}$

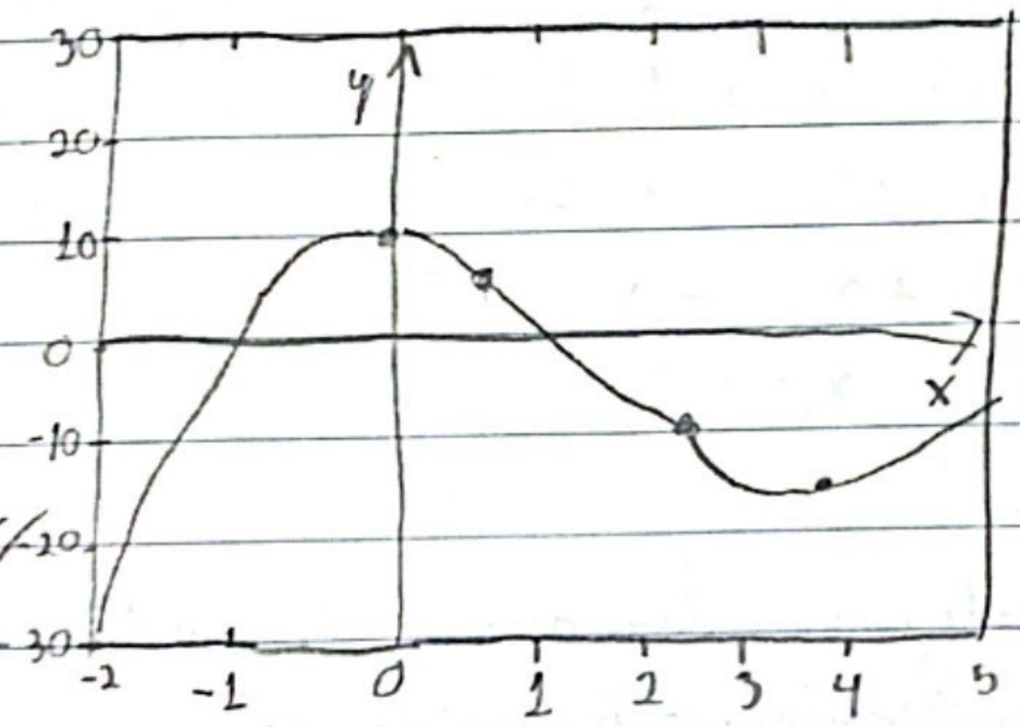
$x_1 = 8 - 3x_3$

$x_1 = 3 - 3x_3$

$x_3 = 7$ //

4ª questão

4- Determine os coeficientes a, b, c e d $p(x) = ax^3 + bx^2 + cx + d$, cujo gráfico passa pelos pontos $P_1 = (0, 10)$, $P_2 = (1, 7)$, $P_3 = (3, -11)$ e $P_4 = (4, -14)$



Coeficiente: $\begin{cases} a = 1 \\ b = -6 \\ c = 2 \\ d = 10 \end{cases}$ //

$\begin{cases} d = 10 \\ a + b + c + d = 7 \\ 27a + 9b + 3c + d = -11 \\ 64a + 16b + 4c + d = -14 \end{cases}$

$S = \begin{bmatrix} 1 & 1 & 1 & -3 \\ 27 & 9 & 3 & -21 \\ 64 & 16 & 4 & -24 \end{bmatrix}$

$-27L_1 + L_2 \to L_2$

$-64L_1 + L_3 \to L_3$

$\frac{1}{4} L_3 \to L_3$

$S = \begin{bmatrix} 1 & 1 & 1 & -3 \\ 0 & -18 & -24 & 60 \\ 0 & -48 & -60 & 168 \end{bmatrix}$

$-\frac{1}{18} L_2 \to L_2$

$S = \begin{bmatrix} 1 & 1 & 1 & -3 \\ 0 & 1 & \frac{4}{3} & -\frac{10}{3} \\ 0 & -48 & -60 & 168 \end{bmatrix}$

$-L_2 + L_1 \to L_1$

$48L_2 + L_3 \to L_3$

$S = \begin{bmatrix} 1 & 0 & -\frac{1}{3} & \frac{1}{3} \\ 0 & 1 & \frac{4}{3} & -\frac{10}{3} \\ 0 & 0 & 4 & 8 \end{bmatrix}$

$S \begin{bmatrix} 1 & 0 & -\frac{1}{3} & \frac{1}{3} \\ 0 & 1 & \frac{4}{3} & -\frac{10}{3} \\ 0 & 0 & 1 & 2 \end{bmatrix}$

$\frac{1}{3} L_3 + L_1 \to L_1$

$-\frac{4}{3} L_3 + L_2 \to L_2$

$S: \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & \frac{4}{3} & -\frac{10}{3} \\ 0 & 0 & 1 & 2 \end{bmatrix}$

$p(x) = x^3 - 6x^2 + 2x + 10$

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5ª questão - Resolva, usando o método de Gauss-Jordan, os seguintes sistemas.

a) $\begin{cases} x_1 + 2x_2 - 3x_4 + x_6 = 2 \\ x_1 + 2x_2 + x_3 - 3x_4 + x_5 + 2x_6 = 3 \\ x_1 + 2x_2 - 3x_4 + 2x_5 + x_6 = 4 \\ 3x_1 + 6x_2 + x_3 - 9x_4 + 4x_5 + 3x_6 = 9 \end{cases}$

$$\left[\begin{array}{cccccc|c} 1 & 2 & 0 & -3 & 1 & 0 & 2 \\ 1 & 2 & 1 & -3 & 1 & 2 & 3 \\ 1 & 2 & 0 & -3 & 2 & 1 & 4 \\ 3 & 6 & 1 & -9 & 4 & 3 & 9 \end{array}\right] \begin{array}{l} \times(-1) \\ L_2 - 1 \cdot L_1 \to L_2 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 2 & 0 & -3 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 2 & 1 \\ 1 & 2 & 0 & -3 & 2 & 1 & 4 \\ 3 & 6 & 1 & -9 & 4 & 3 & 9 \end{array}\right] \begin{array}{l} \times(-1) \\ L_3 - 1 \cdot L_1 \to L_3 \end{array} \qquad \left[\begin{array}{cccccc|c} 1 & 2 & 0 & -3 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 \\ 3 & 6 & 1 & -9 & 4 & 3 & 9 \end{array}\right] \begin{array}{l} \times(-3) \\ L_4 - 3 \cdot L_1 \to L_4 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 2 & 0 & -3 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & 1 & 3 & 3 \end{array}\right] \begin{array}{l} \times(-1) \; L_4 - 1 \cdot L_3 \to L_4 \end{array} \qquad \left[\begin{array}{cccccc|c} 1 & 2 & 0 & -3 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 \end{array}\right] \begin{array}{l} L_4 - 1 \cdot L_3 \to L_4 \\ \times(-1) \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 2 & 0 & -3 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \times(-1) \\ L_1 - 1 \cdot L_3 \to L_1 \end{array} \qquad \left[\begin{array}{cccccc|c} 1 & 2 & 0 & -3 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{cases} x_1 + 2 \cdot x_2 - 3 \cdot x_4 - x_6 = 0 \\ x_3 + 2 \cdot x_6 = 1 \\ x_5 + x_6 = 2 \end{cases} \qquad \begin{cases} x_1 = -2x_2 + 3x_4 + x_6 \\ x_2 = x_2 \\ x_3 = 1 - 2x_6 \\ x_4 = x_4 \\ x_5 = 2 - x_6 \\ x_6 = x_6 \end{cases}$$

$$X = \begin{pmatrix} -2x_2 + 3x_4 + x_6 \\ x_2 \\ 1 - 2x_6 \\ x_4 \\ 2 - x_6 \\ x_6 \end{pmatrix}$$

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b) $\begin{cases} x_1 + 3x_2 - 2x_3 + 2x_5 = 0 \\ 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1 \\ 5x_3 + 10x_4 + 15x_6 = 5 \\ 2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6 \end{cases}$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{array}\right] \begin{array}{l} \times(-1) \\ \\ \\ L_4 - 2L_1 \to L_4 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & -2 & 0 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{array}\right] \begin{array}{l} \\ \times(-1) \\ \\ L_4 - 2L_1 \to L_4 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 10 & 6 \end{array}\right] \begin{array}{l} L_4 - 2L_1 \to L_1 \\ \\ \\ \\ \end{array}$$

$$\times(-5) \quad L_3 - 5L_1 \to L_3$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 8 & 0 & 10 & 6 \end{array}\right] \begin{array}{l} \\ \times(-4) \\ \\ L_4 - 4L_2 \to L_4 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{array}\right] \begin{array}{l} \\ \\ L_4 \leftrightarrow L_3 \\ \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \times\left(\frac{1}{6}\right) \\ L_3/6 \to L_3 \\ \\ \\ \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ \\ \times(-3) \\ L_2 - 3L_3 \to L_2 \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \times(-2) \\ \\ L_1 + 2L_3 \to L_1 \\ \\ \end{array}$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$x_1 = -3x_2 - 4x_4 - 2x_5$ // $x_2 = x_2$ // $x_3 = -2x_4$ // $x_4 = x_4$

$x_5 = x_5$ // $x_6 = \frac{1}{3}$

6ª questão - Sabendo que $\det(A) = -3$ encontre:

a) $\det(A^2) = 9$

b) $\det(A^3) = -27$

c) $\det(A^{-1}) = -\frac{1}{3}$

d) $\det(A^T) = -3$

A) $\det(AB) = \det(A) \cdot \det(B)$

$\det(A^2) = \det(A \cdot A) = \det(A) \cdot \det(A)$

$(-3) \cdot (-3) = 9$

B) $\det(A^3) = \det(A \cdot A^2) =$

$\det(A^3) = (-3) \cdot 9 = -27$

B) $\det(B^{-1}) = \frac{1}{\det(B)}$

$\det(A^{-1}) = \frac{1}{\det(A)} = -\frac{1}{3}$

c) $\det(B) = \det(B^{-1})$

$\det(A^{-1}) = \det(A) = -3$

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7ª questão - Calcule o determinante das seguintes matrizes

a) $\begin{bmatrix} e^{rt} & te^{rt} \\ re^{rt} & (1+rt)e^{rt} \end{bmatrix}$

$\Downarrow$

$\begin{vmatrix} e^{rt} & te^{rt} \\ e^{rt} & (1+rt)e^{rt} \end{vmatrix}$

$\parallel$

$e^{rt}\cdot(1+rt)e^{rt} - te^{rt}\cdot re^{rt}$

$\parallel$

$e^{rt}\cdot(e^{rt}+rte^{rt}) - rte^{2rt} = e^{2rt} + rte^{2rt} - rte^{2rt} = e^{2rt}$ //

b) $\begin{bmatrix} \cos\beta t & \operatorname{sen}\beta t \\ a\cos\beta t - \beta\operatorname{sen}\beta t & a\operatorname{sen}\beta t + \beta\cos\beta t \end{bmatrix}$

$\cos\beta t\cdot(a\operatorname{sen}\beta t + \beta\cos\beta t) - \operatorname{sen}\beta t\cdot(a\cos\beta t - \beta\operatorname{sen}\beta t)$

$a\operatorname{sen}\beta t\cos\beta t + \beta\cos\beta t\cos\beta t - a\operatorname{sen}\beta t\cos\beta t + \beta\operatorname{sen}\beta t\operatorname{sen}\beta t$

$\beta\cos\beta t\cos\beta t + \beta\operatorname{sen}\beta t\operatorname{sen}\beta t$

$\beta(\cos\beta t)^2 + \beta(\operatorname{sen}\beta t)^2$

$\cancel{\cos\beta t\operatorname{sen}\beta t} + \beta\cos^2\beta t - \cancel{\cos\beta t\operatorname{sen}\beta t} + \beta\operatorname{sen}^2\beta t$

$\beta\cos^2\beta t + \beta\operatorname{sen}^2\beta t = \beta(\cos^2\beta t + \operatorname{sen}^2\beta t) = \beta$ //

8ª questão

a) $\begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{vmatrix}$ $\begin{matrix} -5L_1+L_2\to L_2 \\ L_1+L_3\to L_3 \\ -2L_1+L_4\to L_4 \end{matrix}$ $A=\begin{bmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 12 & 0 & -1 \end{bmatrix}$ $-12L_2+L_4\to L_4$ $A=\begin{bmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 108 & 23 \end{bmatrix}$

$-\frac{1}{3}L_3\to L_3$ $A=\begin{bmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 108 & 23 \end{bmatrix}$ $\det(A) = -3\det(A)$ $-108L_3+L_4\to L_4$ $A=\begin{bmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & -13 \end{bmatrix}$ $\det(A) = 39$ //

b) $\begin{vmatrix} 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix}$ $L_2-(\frac{1}{2})\cdot 1\cdot L_1\to L_2$ $\begin{vmatrix} 2 & 1 & 3 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix}$ $L_3-(-4)L_2\to L_2$ $\begin{vmatrix} 2 & 1 & 3 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 & 2 \\ 0 & 1 & 2 & 3 \end{vmatrix}$ $L_4-(-2)$ $L_2\to L_4$ $\begin{vmatrix} 2 & 1 & 3 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 1 & 4 \end{vmatrix}$

$\begin{vmatrix} 2 & 1 & 3 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 6 \end{vmatrix} = \begin{vmatrix} 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 2 & 1 & 3 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 6 \end{vmatrix} = R = 2\cdot\left(\frac{-1}{2}\right)\cdot(-1)\cdot 6 = 6$ //

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c) $\begin{vmatrix} 5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -2 & 0 & 0 & 0 \end{vmatrix}$ $L_5 \leftrightarrow L_2$ $\begin{bmatrix} 5 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -4 \end{bmatrix}$

$$\begin{bmatrix} 5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -2 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -4 \end{bmatrix} = 5 \cdot (-2) \cdot 3 \cdot 1 \cdot (-4) = -120$$

Determine todos os valores de λ para os quais $\det(A - \lambda I_n) = 0$, em que

a) $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 3 & 0 \\ 3 & 2 & -2 \end{bmatrix}$ $P(\lambda) = \det \begin{bmatrix} 1-\lambda & 0 & 0 \\ -1 & 3-\lambda & 0 \\ 3 & 2 & -2-\lambda \end{bmatrix}$

$P(\lambda) = (1-\lambda)(3-\lambda)(-2-\lambda) = 0$

$S = (1, 3, -2)$

$\lambda_1 = 1$

$\lambda_2 = 3$

$\lambda_3 = -2$

b) $\begin{bmatrix} 2 & 2 & 3 \\ 1 & 2 & 1 \\ 2 & 2 & 1 \end{bmatrix}$ $C_2 = 1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\lambda L_3 = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$ $A - \lambda L_3 = \begin{bmatrix} 2 & 2 & 3 \\ 1 & 2 & 1 \\ 2 & 2 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$

$\begin{bmatrix} 2-\lambda & 2 & 3 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 1-\lambda \end{bmatrix}$ $(2-\lambda) \cdot (2-\lambda) - (1-\lambda) + 4 - 6 - 3(2-\lambda) \cdot$

$C_1 - 2 \cdot 1 \cdot (1-\lambda) - (2-\lambda) \cdot (-2) \cdot 2 =$

$\lambda_1 = -1$

$\lambda_2 = 2$

$\lambda_3 = 4$

$\lambda^3 + 5\lambda - 2\lambda + 8$

$-\lambda^3 + 5\lambda^2 - 2\lambda + 8 = 0$

$(1+\lambda) \cdot (-2+\lambda) \cdot (4+\lambda) = 0$

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9 questão - Encontre os valores de a para os quais a matriz $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 2 & a \end{bmatrix}$ tem inversa.

$$\det A \neq 0 = \det \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 2 & a \end{bmatrix} \neq a$$

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 2 & a \end{vmatrix} \begin{vmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 2 \end{vmatrix} \quad 1 \cdot 0 \cdot a + 1 \cdot 0 \cdot 1 + 0 \cdot 1 \cdot 2 - 1 \cdot 0 \cdot 0 - 2 \cdot 0 \cdot 0 - 2 \cdot a \cdot 1 \neq 0$$

$$0 + 0 + 0 - 0 - 0 - a \neq 0$$

$$-a \neq 0$$

$$a \neq 0$$

10ª questão - Se possível, encontre as inversas das seguintes matrizes:

a) $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a+2d+3g & b+2e+3h & c+2f+3i \\ a+d+2g & b+e+2h & c+f+2i \\ d+2g & e+2h & f+2i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$a + 2 - 3g = 1 \quad b + 2e + 3h = 0 \quad c + 2f + 3i = 0$

$a + d + 2g = 0 \quad b + e + 2h = 1 \quad c + f + 2i = 0$

$d + 2g = 0 \quad e + 2h = 0 \quad f + 2i = 1$

$\begin{cases} a = 0 \,//\, b = 1 \quad c = 1 \\ d = 2 \quad e = -2 \quad f = -1 \\ g = -1 \quad h = 1 \quad i = 1 \end{cases} = A^{-1} = \begin{bmatrix} 0 & 1 & -1 \\ 2 & -2 & -1 \\ -1 & 1 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 2 & 2 \\ 1 & 3 & 1 \\ 1 & 3 & 2 \end{bmatrix} \cdot \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a+2d+2g & b+2e+2h & c+2f+2i \\ a+3d+g & b+3e+h & c+3f+i \\ a+3d+2g & b+3e+2h & c+3f+2i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$a + 2d + 2g = 1 \quad b + 2e + 2h = 0 \quad c + 2f + 2i = 0$

$a + 3d + 2g = 0 \quad b + 3e + h = 1 \quad c + 3f + i = 0$

$a + 3d + 2g = 0 \quad b + 3e + 2h = 0 \quad c + 3f + 2i = 1$

$\begin{cases} a = 3 \,//\, b = -2 \,//\, c = -4 \\ d = -1 \,//\, e = 0 \,//\, f = 1 \\ g = 0 \,//\, h = -1 \,//\, i = 1 \end{cases} = B = \begin{bmatrix} 3 & -2 & -4 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & 2 \\ 1 & -1 & 2 & 1 \\ 1 & 3 & 3 & 2 \end{bmatrix} \cdot \begin{vmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} a+e+i+m & b+f+j+n \\ a+2e-i+2m & b+2f-j+2n \\ a-e+2i+m & b-f+2j+n \\ a+3e+3i+2m & b+3f+3j+2n \end{vmatrix}$

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$$\left.\begin{array}{l} c+g+k=0 \quad d+h+l+p \\ c+2g-k+2o \quad d+2h-l+2p \\ c-g+2k+o \quad d-h+2l+p \\ c+3g+2o \quad d+3h+3l+2p \end{array}\right| = \left|\begin{array}{l} 1\ 0\ 0\ 0 \\ 0\ 1\ 0\ 0 \\ 0\ 0\ 1\ 0 \\ 0\ 0\ 0\ 1 \end{array}\right| \quad \begin{array}{l} a=7/3 \quad b=-1/3 \quad c=-1/3 \quad l=-4/9 \\ b=-1/9 \quad g=-4/9 \quad h=1/9 \quad j=-1/9 \\ j=-2/9 \quad k=1/9 \quad l=2/9 \\ m=-5/3 \quad n=2/3 \quad o=2/3 \quad p=1/3 \end{array} \quad = C^{-1} \left|\begin{array}{l} 7/3\ -1/3\ -1/3\ -2/3 \\ 4/9\ -1/9\ -4/9\ 1/9 \\ -1/9\ -2/9\ 1/9\ 2/9 \\ -5/3\ 2/3\ 2/3\ 1/3 \end{array}\right|$$

d)
$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 1 & 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{lll} a+2d+3g & b+2e+3h & c+2f+3i \\ 2d+3g & 2e+3h & 2f+3i \\ a+2d+4g & b+2e+4h & c+2f+4i \end{array} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$a+2d+3g=1 \quad b+2e+3h=0 \quad c+2f+3i=0$

$2d+3g=0 \quad 2e+3h=1 \quad 2f+3i=0$

$a+2d+4g=0 \quad b+2e+4h=0 \quad c+2f+4i=1$

$$\left\{\begin{array}{l} a=1 \quad b=-2 \quad c=0 \\ d=3/2 \quad e=1/2 \quad f=-3/2 \\ g=-1 \quad h=0 \quad i=1 \end{array}\right. \quad D^{-1} = \left|\begin{array}{l} 1\ -2\ 0 \\ 3/2\ 1/2\ -3/2 \\ -1\ 0\ 1 \end{array}\right|$$

e)
$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{lll} a+2d+3g & b+2e+3h & c+2f+3i \\ a+d+2g & b+e+2h & c+f+2i \\ d+g & e+h & f+i \end{array} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$a+2d+3g=1 \quad b+2e+3h=0 \quad c+2f+3i=0 \qquad a=1 \,//\, b=0 \quad c=-2$

$a+d+2g=0 \quad b+e+2h=1 \quad c+f+2i=0 \qquad d=0 \,//\, e=0 \,//\, f=1$

$d+g=0 \quad e+h=0 \quad f+i=1 \qquad g=-1 \quad h=1 \quad i=1$

$$E^{-1} = \left|\begin{array}{l} 1\ 0\ -3 \\ 0\ 0\ 1 \\ -1\ 1\ 1 \end{array}\right| //$$

f)
$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 1 & 2 \\ 1 & 2 & -1 & 1 \\ 5 & 9 & 1 & 6 \end{bmatrix} \cdot \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} a+e+i+m=1 \quad b+f+j+n=0 \quad c+g+k+o=0 \quad d \\ a+3e+i+2m=0 \quad b+3f+j+2n \\ a+2e-i+m=0 \quad b+2f-j+n=0 \quad c \\ 5a+9e+i+6m=0 \quad 5b+9f+j+6n \end{array}$$

$a=1/4 \,//\, e=5/4 \,//\, c=-3/4 \,//\, d=1/2$

$l=1/2 \,//\, f=-1/2 \,//\, g=1/2 \,//\, h=0$

$i=1/4 \,//\, j=1/4 \,//\, k=1/4 \,//\, l=-1/2$

$m=0 \,//\, n=-2 \,//\, o=-2 \,//\, p=-2$

$$F^{-1} = \left|\begin{array}{l} 1/4\ 5/4\ -3/4\ 1/2 \\ 1/2\ -1/2\ 1/2\ 0 \\ 1/4\ 1/4\ 1/4\ -1/2 \\ 0\ -2\ -2\ -2 \end{array}\right| //$$

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